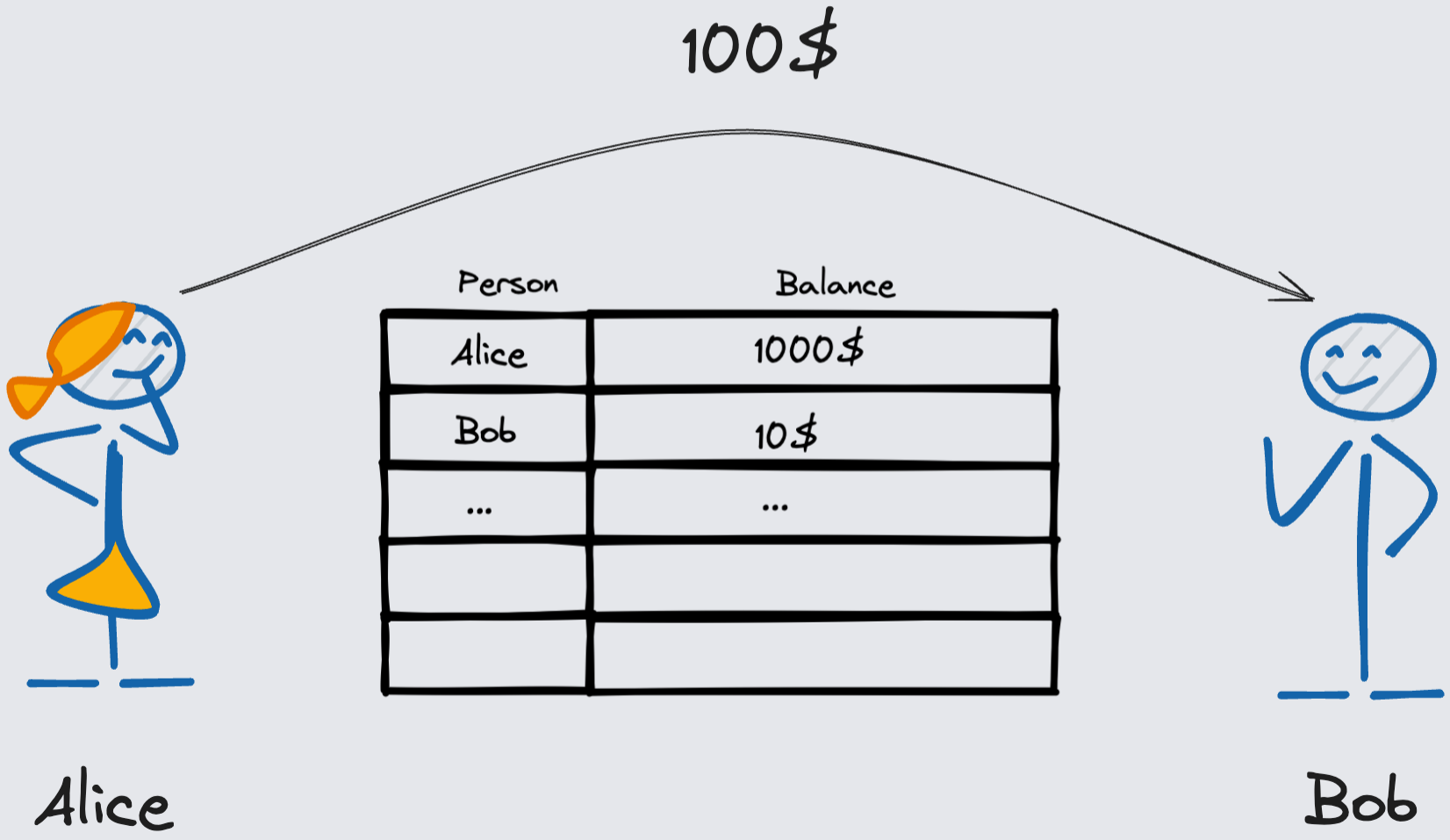
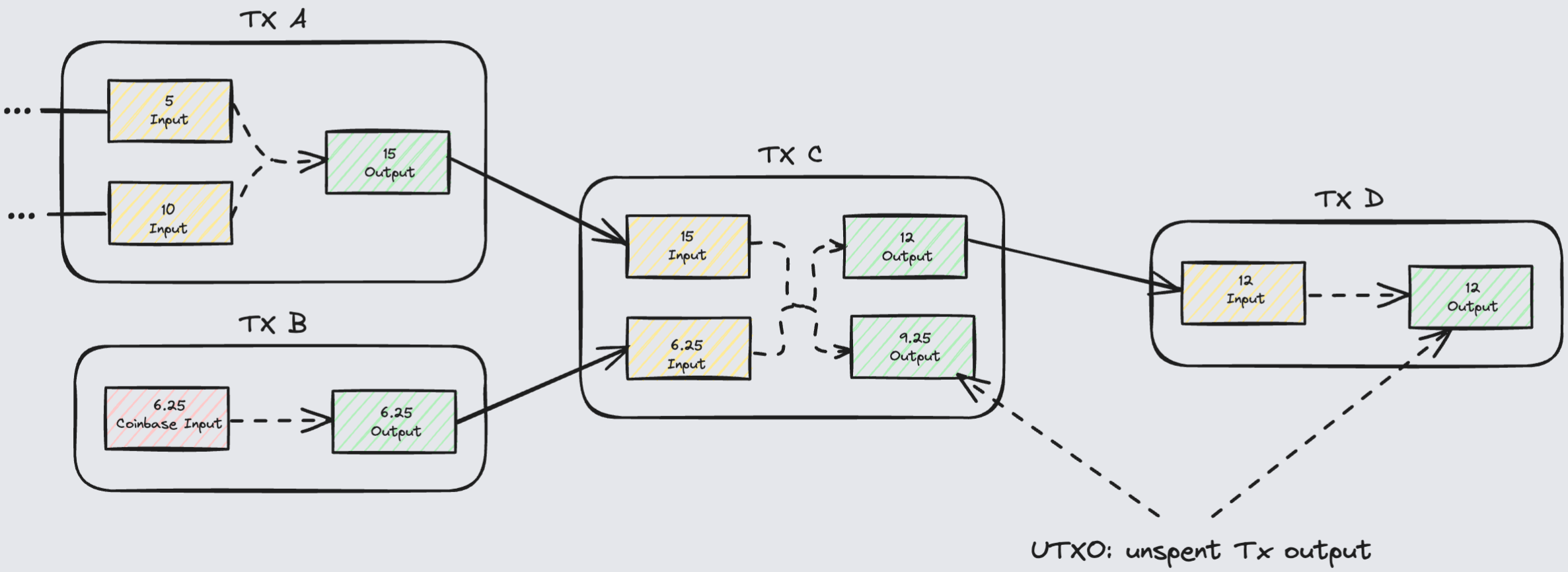
Railgun

Prepared by Kirill Sizov

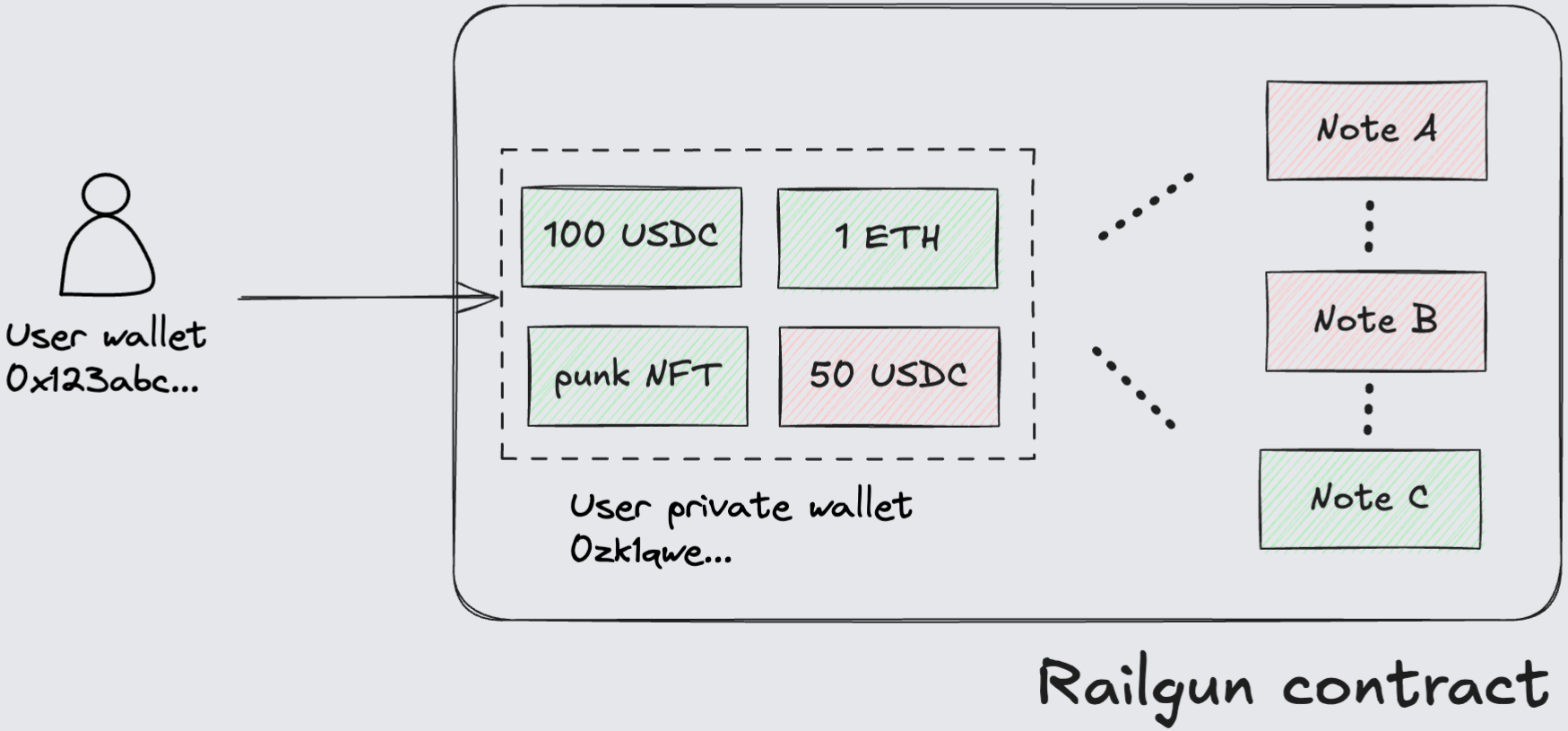
Account-based model (like Ethereum)



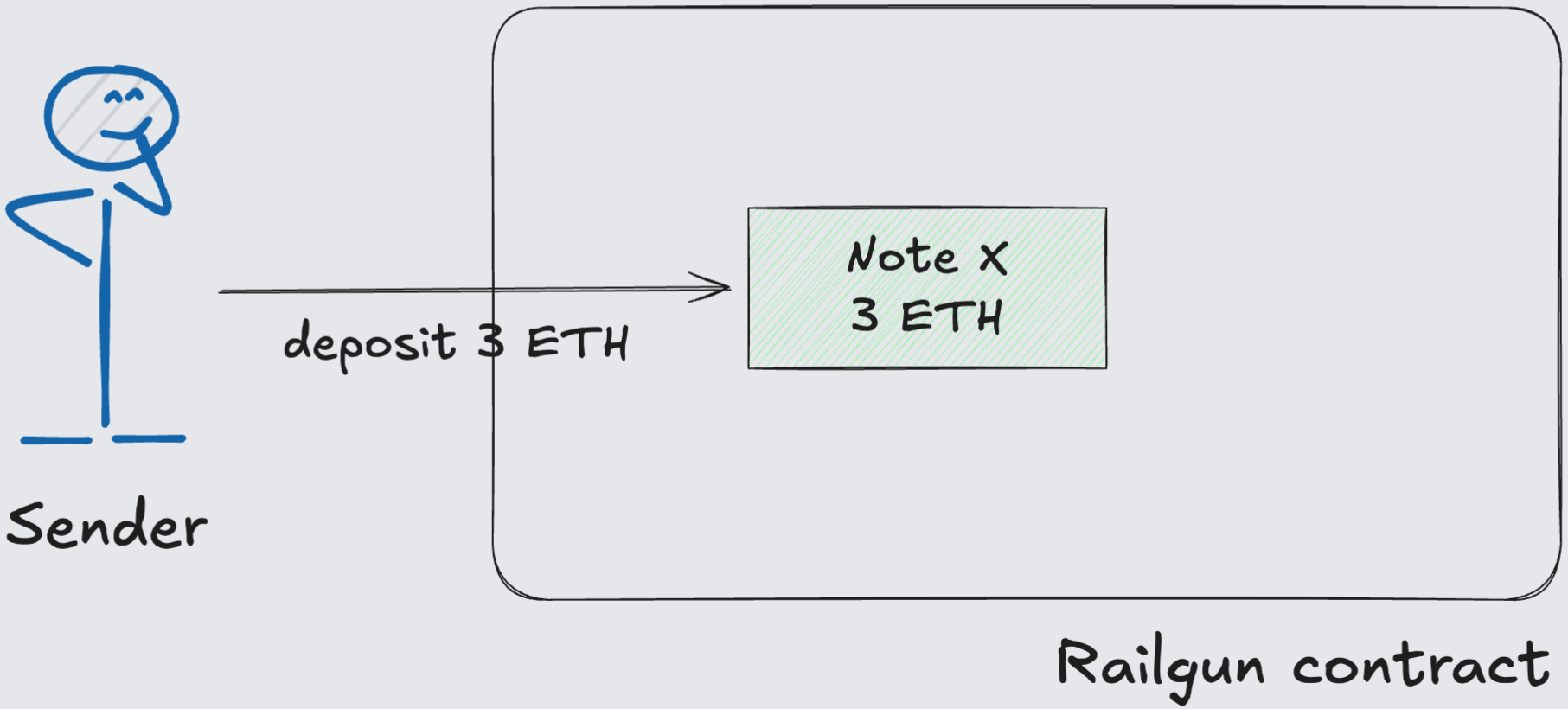
UTXO model (like Bitcoin)



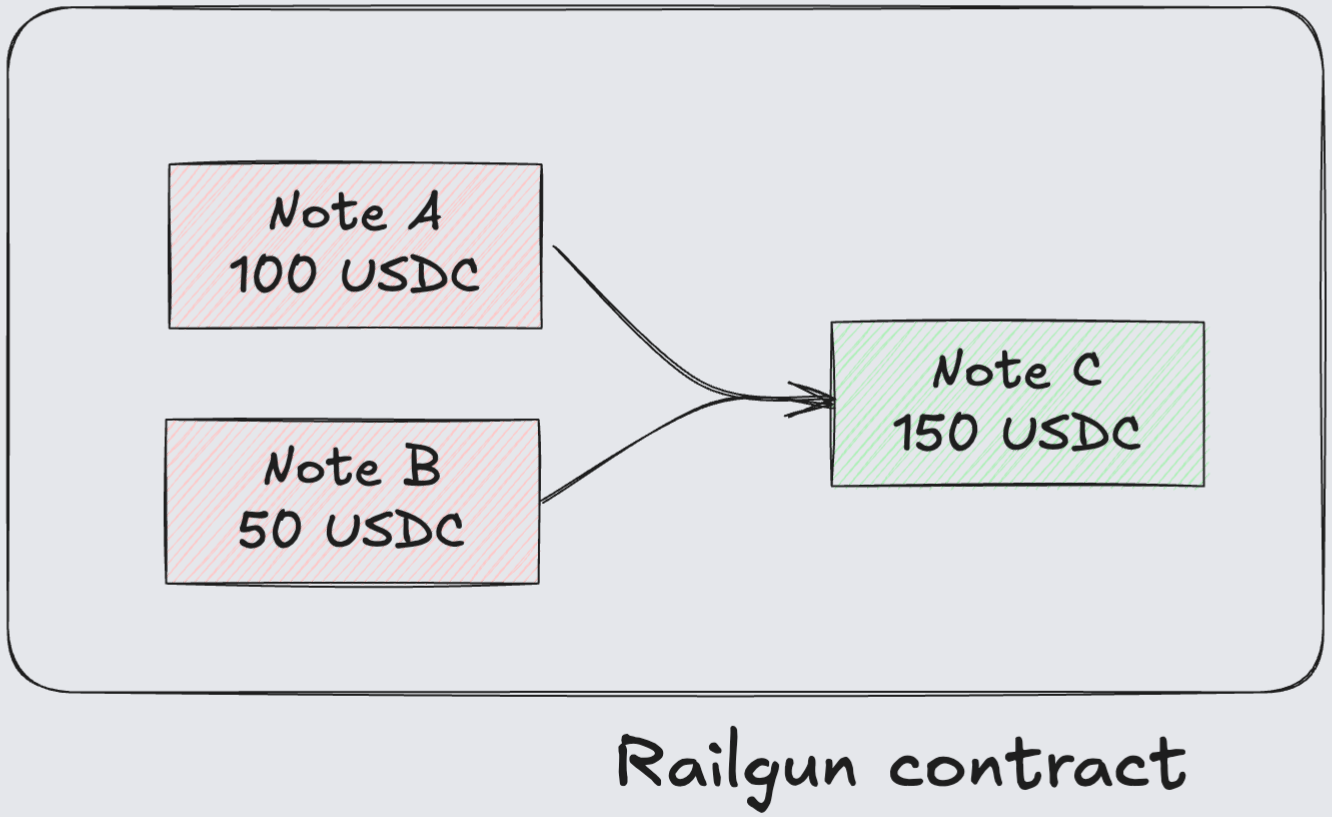
What is railgun



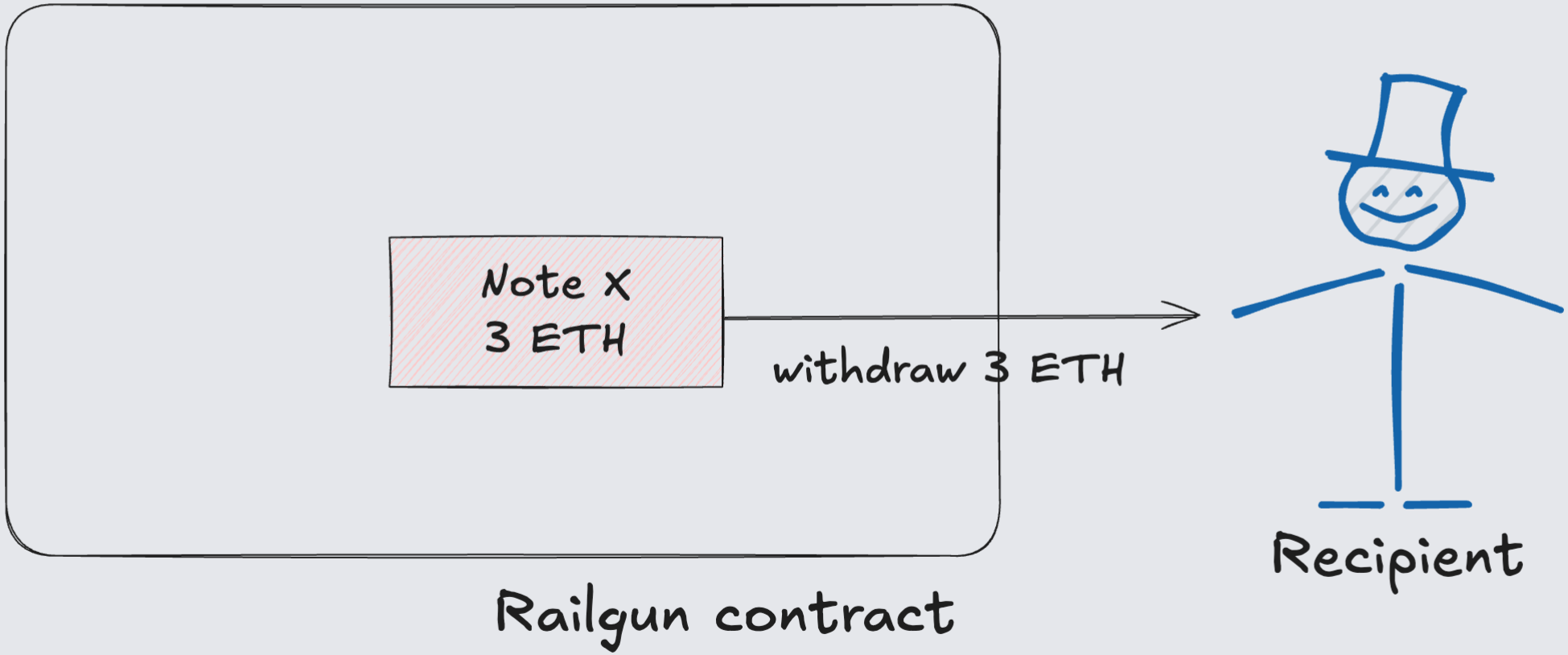
Shield (public → private)



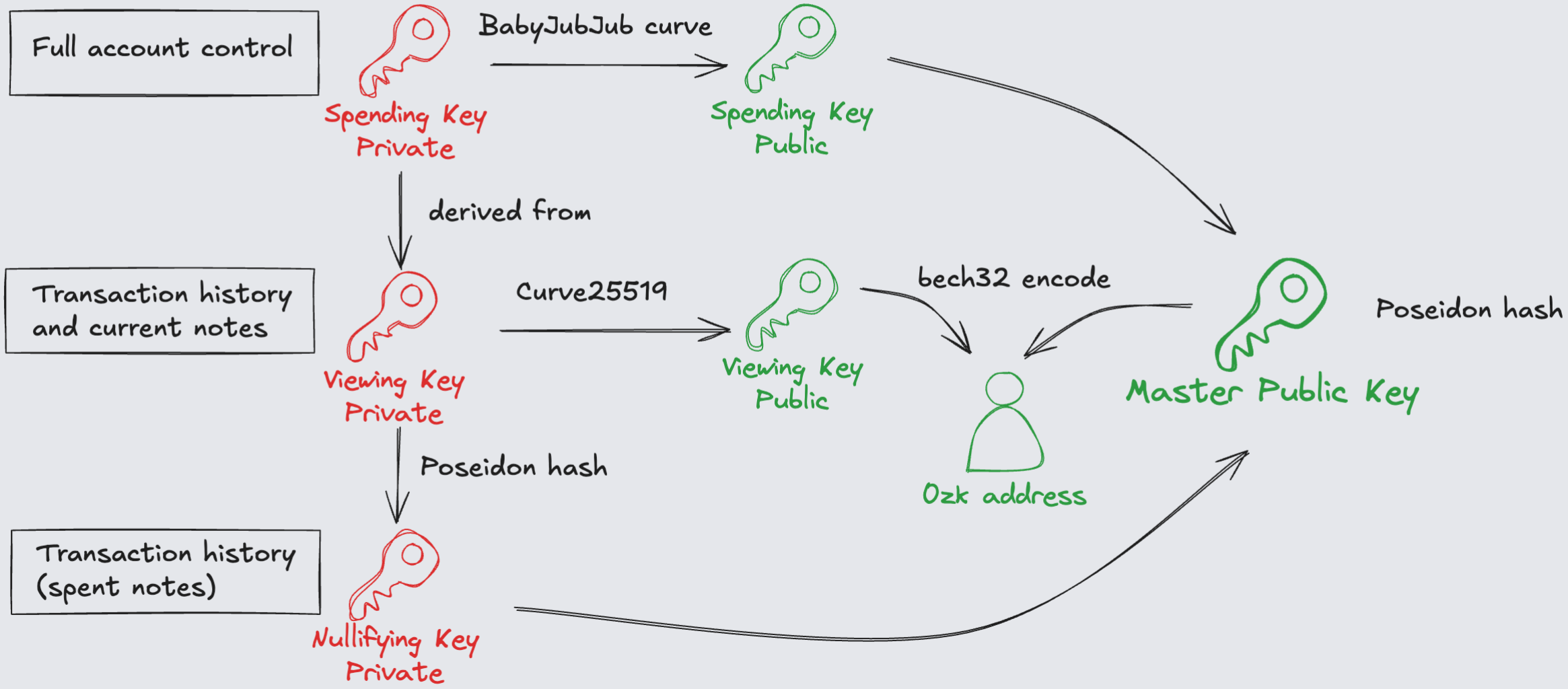
Transact (private → private)



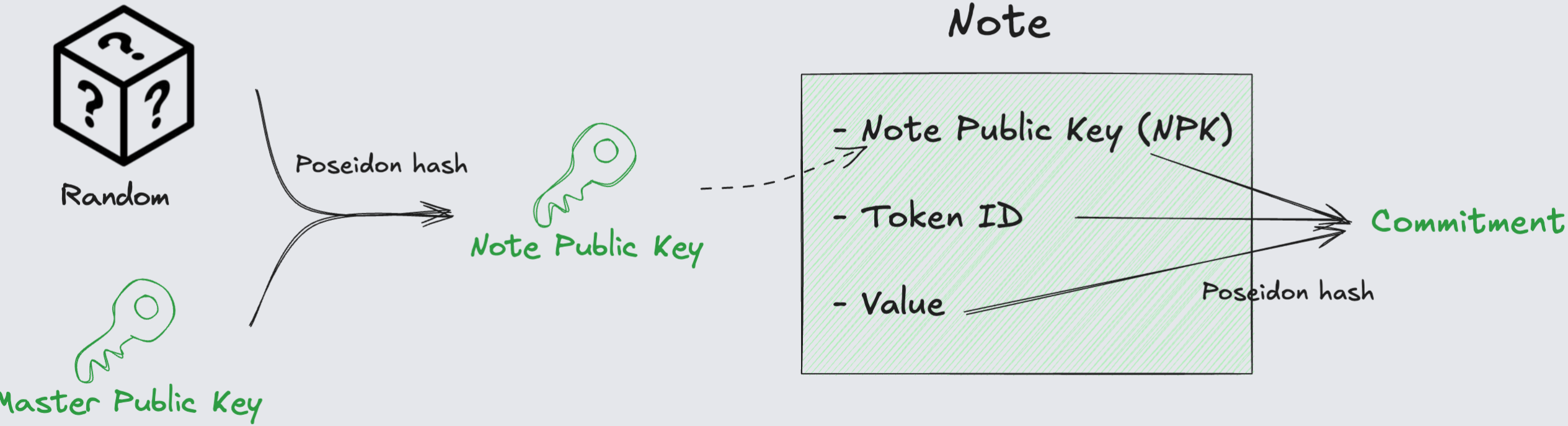
Unshield (private → public)



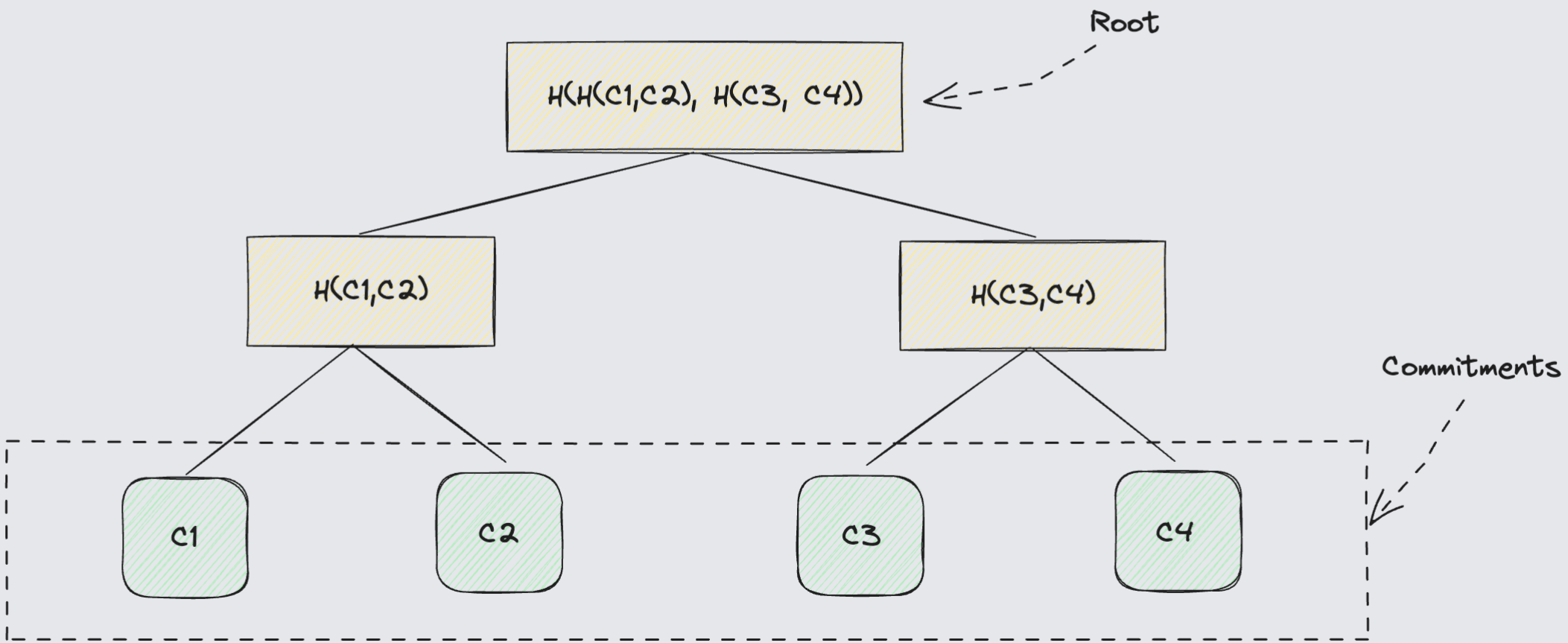
Account keys



Note struct


Random
Master Public Key
Poseidon hash
Note Public Key
Note
- Note Public Key (NPK)
- Token ID
- Value
Poseidon hash
Commitment

Merkle tree



Note spending

Private

General
- Merkle root
- Nullifying key
- Token
- EdDSA signature

For each spending note:
- Leaf index
- Merkle path
- Value
- Random

For each new note:
- Note Public Key
- Value

Public

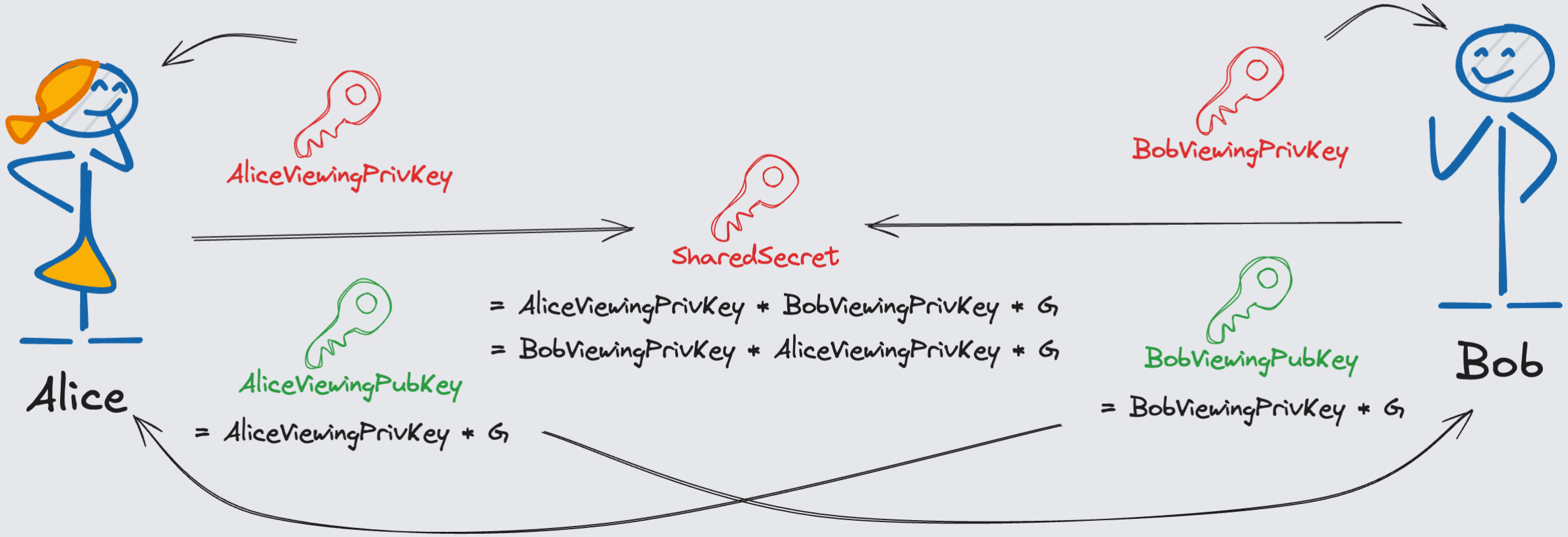
General
- ZK proof (A, B, C)

For each spending note
- Nullifier = Poseidon(nullifyingKey, leafIndex)

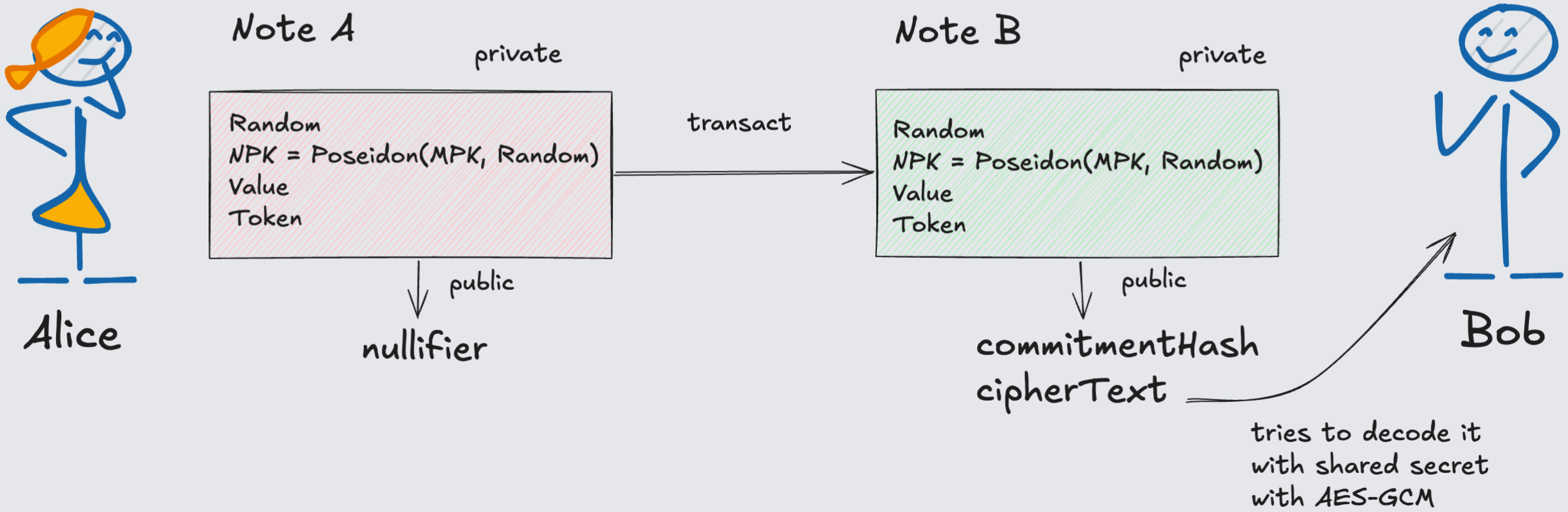
For each new note
- Commitments

How does receiver determines their notes?

ECDH


AliceViewingPrivKey
BobViewingPrivKey
SharedSecret
= AliceViewingPrivKey * BobViewingPrivKey * G
= BobViewingPrivKey * AliceViewingPrivKey * G
Alice
Bob
AliceViewingPubKey
= AliceViewingPrivKey * G
BobViewingPubKey
= BobViewingPrivKey * G

Encryption


Alice
Note A
private
Random
NPK = Poseidon(MPK, Random)
Value
Token
public
nullifier
transact
Note B
private
Random
NPK = Poseidon(MPK, Random)
Value
Token
public
commitmentHash
cipherText
Bob
tries to decode it
with shared secret
with AES-GCM